

Nanyang Technological University
School of Social Sciences

HE3002 Macroeconomics III

Assignment 2

The submission dateline is **7 November 2024** at **1159pm**. Late submission will not be graded at all. Please submit your assignment by uploading it on the Discussion Board. Please submit the hardcopy of your assignment to me in the box placed outside my office at the School of Social Sciences - Level 4 room 66 by 8 November 2024.

1. This question refers to the IS-MPR-PC model described by the IS curve:

$$y_t = y_t^* - \alpha(i_t - \pi_t - r^*) + \epsilon_t^y \quad (1)$$

monetary policy rule:

$$i_t = r^* + \pi^* + \beta_\pi(\pi_t - \pi^*) \quad (2)$$

Phillips curve:

$$\pi_t = \pi_t^e + \gamma(y_t - y_t^*) + \epsilon_t^\pi \quad (3)$$

where y_t is the actual output, y_t^* is the potential output, i_t is the nominal interest rate, π_t is the rate of inflation, r^* is the real interest rate when $y_t = y_t^*$, π^* is the central bank's inflation target and π_t^e is the expected rate of inflation. Assume that α and γ are positive and β_π is larger than 1.

- (a) Show that the solution for inflation is

$$\pi_t = \theta\pi_t^e + (1 - \theta)\pi^* + \theta(\gamma\epsilon_t^y + \epsilon_t^\pi) \quad (4)$$

and

$$\theta = \frac{1}{1 + \alpha\gamma(\beta_\pi - 1)} \quad (5)$$

- (b) Provide intuition to Equation (5). Show that output depends on how far inflation is from the central bank's target.
- (c) What determines whether inflation depends more on the public's expectations or the central bank's target?
- (d) Analyze the effect of an increase in inflation expectation on output and inflation graphically and algebraically. Provide your intuition.

- (e) How does the effect on output and inflation from part (d) depend how aggressive a central bank is in its response to inflation? Explain graphically and algebraically. Provide your intuition.